



CHRONIC KIDNEY
DISEASE
DETECTION

Chronic Kidney Disease Prediction

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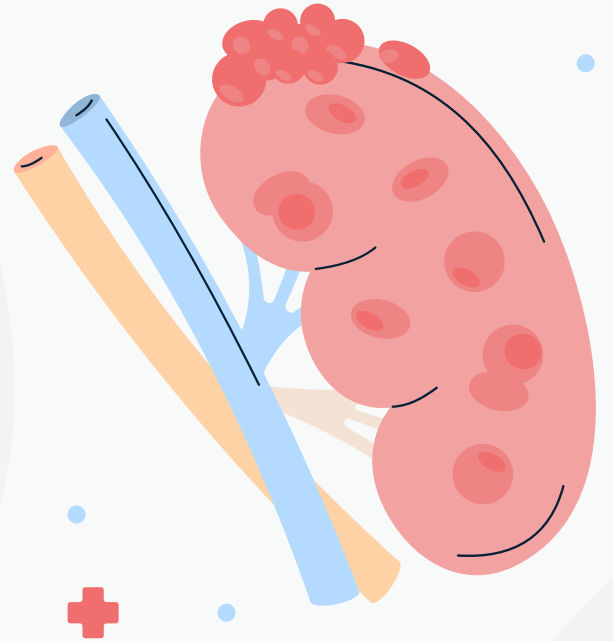


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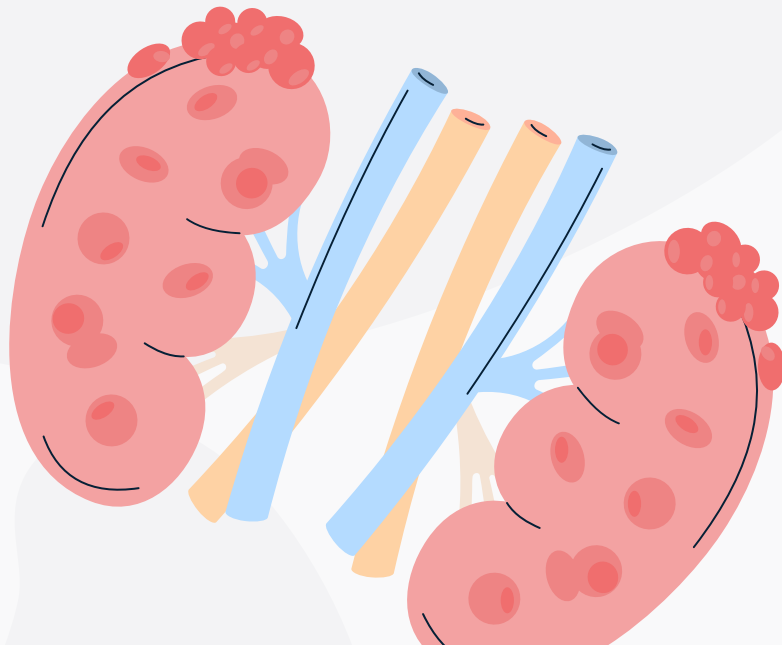
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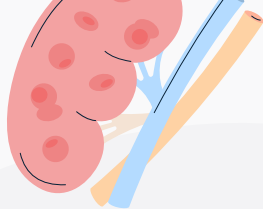
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Introduction



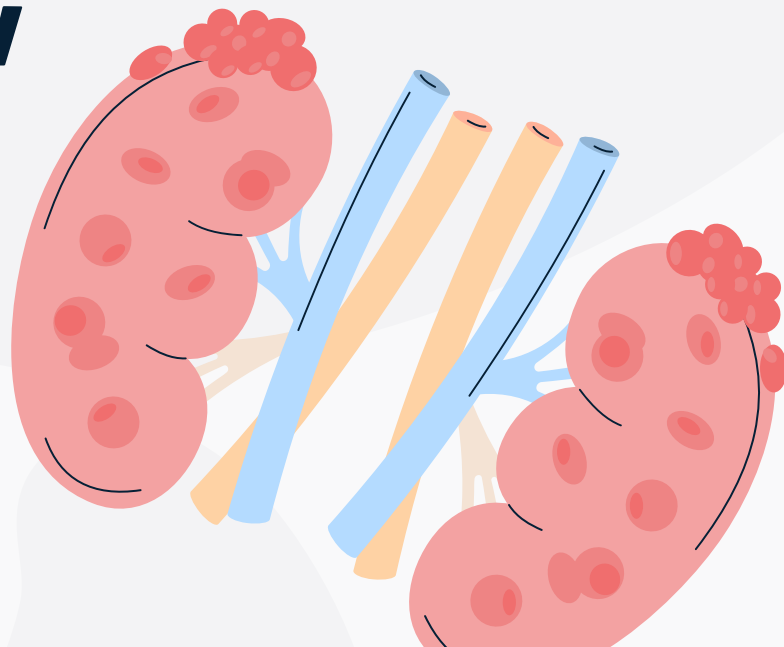


Chronic Kidney Disease

- Chronic Kidney Disease (CKD) is a serious health issue that affects millions worldwide, often progressing silently.
- Early detection plays a vital role in managing the disease and avoiding complications like kidney failure.
- Our project aims to leverage machine learning to predict whether a person has CKD using personal, lifestyle, and clinical data.
- The model helps support early screening efforts and potential clinical decision support tools.

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Data Overview



Data Overview



Data Summary

We used a **Kaggle** dataset with **1,659 records** and **53 features**, covering demographics, health stats, and lifestyle habits



Handling Class Imbalance

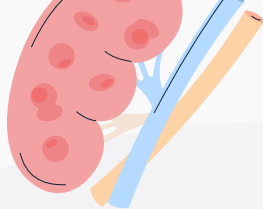
There were fewer **non-CKD cases** than **CKD**, so we used **SMOTE** to fix the imbalance.



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Exploratory Data Analysis (EDA)





Exploratory Data Analysis

- **Understanding the Data**

We explored the distribution of key features. CKD patients tended to have low GFR, high Creatinine, and more protein in their urine

- **Insights & Patterns**

We found correlations between health indicators and CKD. For example, poor diet and low physical activity often appeared with CKD. We used heatmaps and statistical tests to visualize these relationships and guide feature selection.



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Preprocessing & Feature Selection



Preprocessing



Cleaning & Encoding

We dropped some columns that weren't useful for prediction :

- **Education Level**
- **DoctorInCharge**
- **QualityOfLife**
- **SocioEconomicStatus**

We encoded the categorical features with **Standard Scaler** to prepare the dataset for modeling



Feature Selection

We used **Mutual Information**, **Random Forest Classifier**, **Recursive Feature Elimination** to select the most important features. These helped us focus on strong indicators like **GFR**, **Creatinine**, and **Protein in Urine**. Using **correlation** and **chi-squared** tests, we picked key features like **GFR** and **Creatinine** for prediction.

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Machine Learning Models



Machine Learning Models

Model Training

We trained multiple models including :

- Logistic Regression
- Random Forest
- Decision Tree
- XGBoost

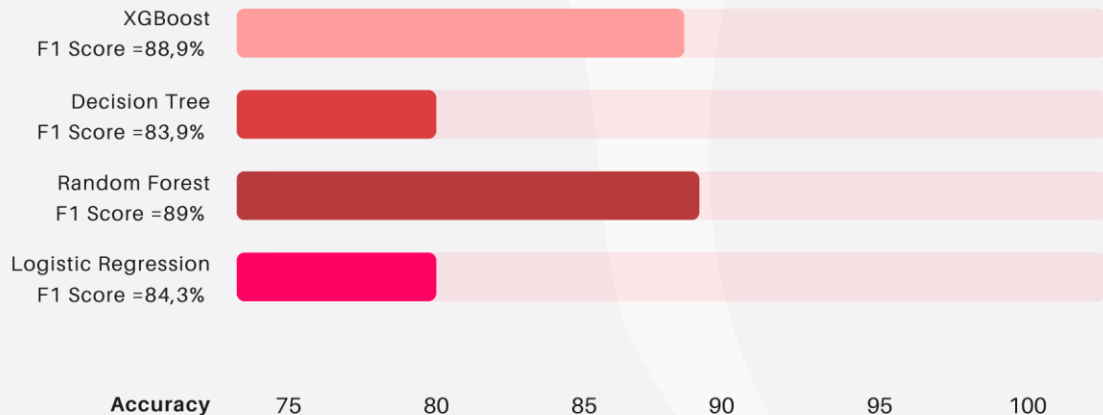
to compare their performance.

Evaluation Setup

The data was split into training and testing sets. We used metrics like accuracy and F1-score to evaluate which model worked best

Model Evaluation

Model Evaluation



Choosing the Model

Although **Random Forest** had slightly higher accuracy, **XGBoost** provided a more suitable and balanced performance across all metrics (PrePrecision, Recall, F1-Score, ROC-AUC). Making it out final choice



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Deployment



Deployment

To make our model more usable, we built a **Streamlit** app where you can input patient data and get a prediction instantly. It's simple, interactive, and designed so that even non-technical users can benefit from it.

Heart Assessment

Patient Information

Basic Health Metrics

BMI - +

Systolic Blood Pressure (mmHg) - +

Fasting Blood Sugar (mg/dL) - +

Kidney Function Tests

Serum Creatinine (mg/dL) - +

BUN Levels (mg/dL) - +

GFR (mL/min/1.73m²) - +

Protein in Urine 0 3

Electrolytes & Cholesterol

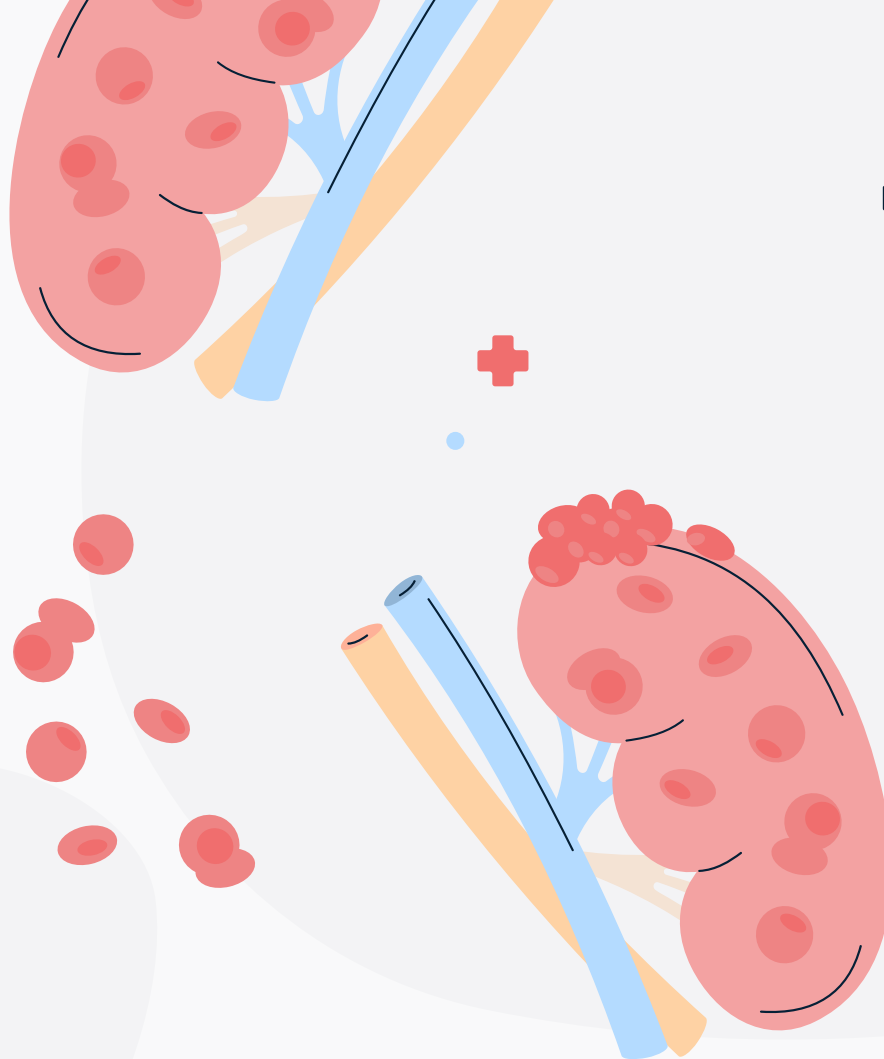
Serum Sodium (mEq/L) - +

Serum Potassium (mEq/L) - +

Serum Phosphorus (mg/dL) - +

HDL Cholesterol (mg/dL) - +

Triglycerides (mg/dL) - +



Thanks

Any questions ?